

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Master of Technology (M.Tech.)

Semester – II University Examination May 2010

M.Tech. (C.S.E.)

EC – 709 Digital Image & Speech Processing

Date: 23.06.10, Wednesday

Time: 12.30 p.m. to 03.30 p.m.

Maximum Marks:70

Instructions:

1. Attempt all questions.
2. Answer to the two sections must be written in the separate answer books.
3. Figures to the right indicate *full* marks.
4. Make suitable assumptions and draw the neat figures wherever required.

SECTION- I

Q.1

- | | | |
|---|---|---|
| A | Explain the concept of Adjacency, Connectivity, Regions and Boundaries. Give suitable examples. | 4 |
| B | What is Contrast stretching & Bit-plane slicing? | 4 |
| C | What is Homomorphic Filtering? | 4 |

Q.2

- | | | |
|---|---|---|
| A | Explain Noise Probability Density Functions | 6 |
| B | What is Pseudocolor Image Processing? | 5 |

OR

- | | | |
|---|---|---|
| A | What is Image Sampling & Quantization? | 6 |
| B | Which are the fields that use Digital Image Processing? Explain any one of them | 5 |

Q.3

- | | | |
|---|---|---|
| A | What is Histogram Processing? | 6 |
| B | Which are the Order-Statistics Filters? | 6 |

OR

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|---|---|---|
| A | Why the Padding is needed in Image Enhancement? | 6 |
| B | Write short note on RGB Color Model | 6 |

SECTION – II

Q.4

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|----------|---|---|
| A | Discuss Typical Applications of Speech Processing | 6 |
| B | Find the Z- transform of the sum of two real exponentials
$X(n) = (1/2)^n u(n) + (1/3)^n u(n)$ | 5 |

Q.5

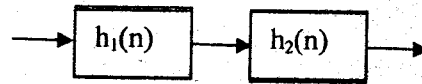
- | | | |
|----------|---|---|
| A | Explain DPCM with adaptive quantization | 6 |
| B | Explain the mechanism of speech production. | 6 |

OR

- | | | |
|----------|--|---|
| A | Explain Pitch Period Estimation using a Parallel Processing Approach | 6 |
| B | Design Filter Banks using IIR Filters | 6 |

Q.6

- | | | |
|----------|--|---|
| A | Compare the Methods of Solution of the LPC Analysis Equations | 6 |
| B | Consider two linear shift invariant systems in cascade i.e. the output of the first is the input to the second | 6 |



Show that the impulse response of the overall system is $h(n) = h_1(n) \otimes h_2(n) = h_2(n) \otimes h_1(n)$

OR

- | | | |
|----------|---|---|
| A | With Reference to Acoustic theory of speech production, explain the terms | 6 |
| | i) Vowels | |
| | ii) Diphthongs | |
| B | Which are the Applications of LPC Parameters? Explain any one of them | 6 |